



Stochastic Processes

Lecture 21

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A Correction



Proof of Lemma 3 (If Part)

- Suppose P is irreducible and every state is positive recurrent
- We will now construct a vector $\mathbf{v}$ satisfying $\mathbf{v} = \mathbf{v}P$
- Fix an arbitrary $x \in \mathcal{X}$, and define

$$V_x(y) := \sum_{t=2}^{\tau_x^{(1)}} \mathbf{1}_{\{X_t=y\}}, \quad y \in \mathcal{X}.$$

- Observe that:
 - $V_x(x) = 1$.
 - $\sum_{y \in \mathcal{X}} V_x(y) = \tau_x^{(1)}$.
- Keeping x fixed, define

$$v_x(y) = \mathbb{E}[V_x(y) \mid \{X_1 = x\}], \quad y \in \mathcal{X},$$

and set $\mathbf{v}_x = [v_x(y) : y \in \mathcal{X}]$

- In the above proof of the existence of a stationary distribution for an irreducible TPM on a finite state space (Lecture 19), the definition of $V_x(y)$ should be

$$V_x(y) = \sum_{t=1}^{\tau_x^{(1)}-1} \mathbf{1}_{\{X_t=y\}}, \quad y \in \mathcal{X}.$$

Ergodicity and Convergence to Stationary Distribution

Theorem (Ergodicity and Convergence to Stationary Distribution)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a homogeneous DTMC on a finite state space $\mathcal{X}$.

Let μ_n denote the PMF of X_n for each n .

If P is ergodic with unique invariant distribution $\pi = [\pi_x : x \in \mathcal{X}]$, then

$$\lim_{n \rightarrow \infty} d_{\text{TV}}(\mu_n, \pi) = \lim_{n \rightarrow \infty} \frac{1}{2} \sum_{x \in \mathcal{X}} |\mu_n(x) - \pi_x| = 0,$$

where d_{TV} denotes the **total variation distance**.

Remarks:

- The proof of ergodicity theorem is based on an important idea called **coupling**
- The above theorem implies that

$$X_n \xrightarrow{\text{d.}} \pi.$$

- **Role of aperiodicity:**

If P is irreducible, aperiodic, and has $|\mathcal{X}| < +\infty$, then there exists $N \in \mathbb{N}$ such that $P^N > \mathbf{0}$

Almost-Sure Convergence in Markov Chains

Almost-Sure Convergence in Markov Chains

Theorem (Almost-Sure Convergence in Markov Chains)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be an **irreducible** DTMC on a finite state space $\mathcal{X}$.
Let π denote the unique stationary distribution of P . Then,

$$\frac{1}{n} \sum_{k=1}^n \mathbf{1}_{\{X_k = \gamma\}} \xrightarrow{\text{a.s.}} \pi_{\gamma} \quad \forall \gamma \in \mathcal{X}.$$

More generally, for any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$,

$$\frac{1}{n} \sum_{k=1}^n f(X_k) \xrightarrow{\text{a.s.}} \sum_{\gamma \in \mathcal{X}} \pi_{\gamma} f(\gamma).$$

Remarks:

- The ergodic theorem is the analogue of SLLN for Markov chains
- The above result does not need aperiodicity

Proof of Theorem 2

Based on Proof of Theorem 4.16 in [LP17]

- Without loss of generality, suppose that $X_1 = x$ for some $x \in \mathcal{X}$
- Setting $\tau_x^{(0)} = 1$, let

$$\tau_x^{(k)} = \inf\{n > \tau_x^{(k-1)} : X_n = x\}, \quad k \in \mathbb{N}.$$

- Because $\mathbf{X}$ is irreducible and has finite state space, every state is positive recurrent
- As a consequence, $\tau_x^{(k)}$ is a stopping time for every k
- Let $\{Y_k\}_{k \in \mathbb{N}}$ be defined as

$$Y_k := \sum_{s=\tau_x^{(k-1)}}^{\tau_x^{(k)}-1} f(X_s).$$

- Using the IID block structure with respect to hitting times, we get that $\{Y_k\}_{k \in \mathbb{N}}$ is an **IID process**
- Furthermore,

$$\mathbb{E}[Y_1 \mid \{X_1 = x\}] = \mathbb{E} \left[\sum_{s=\tau_x^{(0)}}^{\tau_x^{(1)}-1} f(X_s) \mid \{X_1 = x\} \right] \leq f^* \left(\mathbb{E}[\tau_x^{(1)} - \tau_x^{(0)} \mid \{X_1 = x\}] \right), \quad f^* = \max_{x \in \mathcal{X}} f(x).$$

Proof of Theorem 2

Based on Proof of Theorem 4.16 in [LP17]

- Using the IID block structure, we get that

$$\tau_x^{(1)} - \tau_x^{(0)}, \tau_x^{(2)} - \tau_x^{(1)}, \tau_x^{(3)} - \tau_x^{(2)}, \dots$$

are IID random variables with finite mean, and therefore $\mathbb{E}[Y_1 \mid \{X_1 = x\}] < +\infty$

- Using SLLN, we then have

$$\frac{1}{n} \sum_{s=1}^{\tau_x^{(n)}-1} f(X_s) = \frac{1}{n} \sum_{\ell=1}^n Y_\ell \xrightarrow{\text{a.s.}} \mathbb{E}[Y_1 \mid \{X_1 = x\}],$$

$$\frac{\tau_x^{(n)} - 1}{n} = \frac{1}{n} \sum_{\ell=1}^n (\tau_x^{(\ell)} - \tau_x^{(\ell-1)}) \xrightarrow{\text{a.s.}} \mathbb{E}[\tau_x^{(1)} - 1 \mid \{X_1 = x\}].$$

- Combining the above results, we get

$$\frac{\sum_{s=1}^{\tau_x^{(n)}-1} f(X_s)}{\tau_x^{(n)} - 1} \xrightarrow{\text{a.s.}} \frac{\mathbb{E}[Y_1 \mid \{X_1 = x\}]}{\mathbb{E}[\tau_x^{(1)} - 1 \mid \{X_1 = x\}]}.$$

Proof of Theorem 2

Based on Proof of Theorem 4.16 in [LP17]

- Notice that

$$\begin{aligned}
 \mathbb{E}[Y_1 \mid \{X_1 = x\}] &= \mathbb{E} \left[\sum_{s=1}^{\tau_x^{(1)}-1} f(X_s) \mid \{X_1 = x\} \right] = \mathbb{E} \left[\sum_{s=1}^{\tau_x^{(1)}-1} \sum_{\gamma \in \mathcal{X}} f(\gamma) \mathbf{1}_{\{X_s=\gamma\}} \mid \{X_1 = x\} \right] \\
 &= \mathbb{E} \left[\sum_{\gamma \in \mathcal{X}} \sum_{s=1}^{\tau_x^{(1)}-1} f(\gamma) \mathbf{1}_{\{X_s=\gamma\}} \mid \{X_1 = x\} \right] = \sum_{\gamma \in \mathcal{X}} f(\gamma) \mathbb{E} \left[\sum_{s=1}^{\tau_x^{(1)}-1} \mathbf{1}_{\{X_s=\gamma\}} \mid \{X_1 = x\} \right] \\
 &= \sum_{\gamma \in \mathcal{X}} f(\gamma) \mathbb{E} [V_x(\gamma) \mid \{X_1 = x\}] = \sum_{\gamma \in \mathcal{X}} f(\gamma) \frac{\mathbb{E} [V_x(\gamma) \mid \{X_1 = x\}]}{\mathbb{E}[\tau_x^{(1)} - 1]} \mathbb{E}[\tau_x^{(1)} - 1] \\
 &\stackrel{\text{Slide 1}}{=} \sum_{\gamma \in \mathcal{X}} f(\gamma) \pi_\gamma \mathbb{E}[\tau_x^{(1)} - 1] = \mathbb{E}[\tau_x^{(1)} - 1] \left(\sum_{\gamma \in \mathcal{X}} f(\gamma) \pi_\gamma \right).
 \end{aligned}$$

Markov Chain Monte Carlo (MCMC) Techniques

Objective

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To evaluate a sum (integral) of the form

$$I = \sum_{x \in \mathcal{X}} f(x) \pi_x,$$

for given functions $f, \pi : \mathcal{X} \rightarrow \mathbb{R}$, where π is some PMF on $\mathcal{X}$. Here, $\mathcal{X} \subseteq \mathbb{R}^d$ for some d .

Difficulties:

- Direct sampling from π (via ITT or rejection sampling methods) may be difficult

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- Direct sampling from π (via ITT or rejection sampling methods) may be difficult
- High dimensionality: d may be large (e.g., 10000)
- π may only be known up to a scaling factor

Metropolis–Hastings Algorithm

The Metropolis–Hastings MCMC Algorithm

- Originally proposed by Metropolis et al. [MRR⁺53] and Hastings [Has70]
- The key idea: construct an **ergodic Markov chain** whose stationary distribution is π
- If $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ denotes the above Markov chain, it follows from Theorem 2 that

$$\frac{1}{n} \sum_{k=1}^n f(X_k) \xrightarrow{\text{a.s.}} I.$$

- The method also applies when $\mathcal{X}$ is countable or uncountable

Construction of Markov Chain (Given a Target π)

- Let Q be any row-stochastic matrix on $\mathcal{X} \times \mathcal{X}$, i.e.,

$$Q(x, y) > 0, \quad \sum_y Q(x, y) = 1 \quad \forall x, y \in \mathcal{X}.$$

- Let S be a non-negative, **symmetric** matrix having same dimensions as Q
- Let $A = [A(x, y)]_{x, y \in \mathcal{X}}$ be defined via

$$A(x, y) = \frac{S(x, y)}{1 + \frac{\pi_x Q(x, y)}{\pi_y Q(y, x)}}.$$

- The entries of S are chosen in such a way that $0 < A(x, y) \leq 1$ for all $x, y \in \mathcal{X}$
- A is called the **acceptance matrix**

Construction of Markov Chain (Given a Target π)

- Choose an arbitrary starting state $X_1 = x$
- At each time step $n \geq 2$, do the following two tasks:
 - Based on X_{n-1} , sample Y according to

$$\mathbb{P}(\{Y = y\} \mid \{X_{n-1} = x\}) = Q(x, y).$$

- Define X_n as

$$X_n = \begin{cases} y, & \text{with probability } A(x, y), \\ X_{n-1}, & \text{with probability } 1 - A(x, y). \end{cases}$$

- **Claim:** The process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a homogeneous DTMC with π as its stationary distribution

Proof of Claim

- It is left as **exercise** to show that $\mathbf{X}$ is a DTMC and is homogeneous
- For any $x, z \in \mathcal{X}$, we have

$$P(x, z) = \mathbb{P}(\{X_n = z\} \mid \{X_{n-1} = x\}) = \begin{cases} Q(x, z) A(x, z), & z \neq x, \\ 1 - \sum_{y \neq x} Q(x, y) A(x, y), & z = x. \end{cases}$$

- **Exercise:** It is easy to verify that for all $x, z \in \mathcal{X}$,

$$\pi_x P(x, z) = \pi_z P(z, x).$$

The above equations are referred to as **local balance equations** or **reversibility equations**

- Summing over x on both sides, we get

$$\pi_z = \sum_x \pi_x P(x, z),$$

thus proving that π is a stationary distribution of $\mathbf{X}$

Choice of S Matrix in Metropolis–Hastings Algorithm

- The matrix S used in [MRR⁺53] (call this $S^{(M)}$) is given by

$$S^{(M)}(x, y) = \begin{cases} 1 + \frac{\pi_x Q(x, y)}{\pi_y Q(y, x)}, & \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)} \geq 1, \\ 1 + \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)}, & \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)} < 1. \end{cases}$$

With $S = S^{(M)}$ as above, the acceptance matrix A becomes

$$A^{(M)}(x, y) = \begin{cases} 1, & \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)} \geq 1, \\ \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)}, & \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)} < 1 \end{cases} = \min \left\{ 1, \frac{\pi_y Q(y, x)}{\pi_x Q(x, y)} \right\}.$$

- Barker [Bar65] proposes using $S = S^{(B)}$, where $S_{x, y}^{(B)} = 1$ for all $x, y \in \mathcal{X}$

Choice of S Matrix in Metropolis–Hastings Algorithm

- If Q is symmetric (e.g., every row of Q is uniform), then

$$A_{x,y}^{(M)} = \min \left\{ 1, \frac{\pi_y}{\pi_x} \right\}, \quad A_{x,y}^{(B)} = \frac{\pi_y}{\pi_x + \pi_y}.$$

- Thus, if $\pi_x = \pi_y$, then $A^{(M)}(x, y) = 1$, whereas $A^{(B)}(x, y) = 0.5$
- [MRR⁺53] suggests picking $X_{n+1} = y$ with probability 1
- [Bar65] suggests taking $X_{n+1} = X_n = x$ with probability 0.5 and taking $X_{n+1} = y$ with probability 0.5

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